

Update 2 – Weekly Update Report

CSE 499B: Senior Design II | Section: 12 | Group: 03

Week Update Summary

We created a Selenium automation agent and an MCP stdio server that could automatically fill out and submit Google Forms via a structured command-line interface (CLI) workflow using Python. We also created a Software Questions dataset, tokenized it with a GPT-2 tokenizer, split it into training & test sets, and then fine-tuned a pre-trained GPT-2 model with the Hugging Face Trainer API. In addition, we completed my Research Direction: Creating a Weakness-Driven, Fully Offline Adaptive AI Mock Interview System using Open-Source Models. We researched related work to better understand local execution efficiency, quantization, and adaptive logic. Finally, we trained and deployed a personalized technical interview assistant using the DeepSeek-R1 model with LoRA fine-tuning, enabling it to perform AI-driven technical Interviews Locally without using external APIs.

Contribution

Md. Kamrul Islam (2211745642):

This week, I designed and implemented an automation agent using Python, Selenium, and an MCP stdio server. The CLI-based agent prompts the user for a Google Form URL, then triggers a custom MCP tool that launches Google Chrome, navigates to the provided URL, reads the questions in the form, waits for the user's answer, writes the answers to the form, and submits the form by clicking the "Submit" button. The automation flow includes structured task execution, basic confirmation checking, and a clean browser shutdown to ensure stability. After completing each major task step, there is a 5-second pause.

Farhan Ishraque (2212002042):

I rode the waves of Software Questions.csv (dataset), created a text column combining Question and Answer for model input, and then tokenized our data with a GPT-2 tokenizer. I took special care to maintain data integrity by paying attention to sequence length and the pad token during the preprocessing. Finally, I converted the pandas DataFrame into a Hugging Face dataset. After tokenizing the dataset, I split the data into training (80%) and test (20%) sets for model validation. I fine-tuned a pre-trained GPT2 using the HuggingFace Trainer API, providing Training Arguments to set the batch size, learning rate, evaluation strategy, etc., to optimize training. Finally, I saved the fine-tuned model and tokenizer, allowing them to be reused for future inference. Then I read some research papers for the mock interview literature review.

Iram Shehzad (2131614642):

I finalised the research direction for my project and decided to focus on building a weakness-driven adaptive AI mock interview system using an open-source model that doesn't

rely on any external APIs. The system will adjust the difficulty of questions on the fly based on structured evaluation dimensions such as technical accuracy, clarity, and depth. I also looked at a few other works that are similar, such as AI-Based Interview Coach (2025), which uses Vosk for offline transcription and local storage; InterviewX (2024), which focuses on QLoRA and quantised models for efficient local execution; and Smart Prep – Local Variant (2025), which uses sentence embeddings for local semantic matching. I learned more about how to set up fully offline AI systems that work well and how to add adaptive logic without making many changes.

Atikul Islam Nahid (2211978042):

I am equipped to assist in conducting technical interviews as an AI job interview assistant by using the DeepSeek-R1 language model, which was trained on my own technical interview questions and answers, to produce a custom AI assistant for initial phone interviews with candidates. As of today, I have completed implementing my datasets into the model by running the model locally on a machine. In addition, I developed the model using the LoRa Fine-tuning methodology and trained and configured it to respond to questions using the conventional text-generation pipeline. To summarize, I developed a new, personalized language model to conduct technical interviews for job candidates.

Completed Task:

- Various model training with the dataset.
- Read, write, and automatic Google Form submission.

Work in Progress

- Comparing various models' performance.
- Integrating a custom model into the mock interview feature.
- Upload files to a Google Form and submit it automatically.

Next Target

- Integrate the custom model into the mock interview feature.
- Fully read and write on a Google Form and submit it automatically.

Challenges

- Selecting and using the specific models for a specific task.
- Bypassing Google Chrome's automation blocking.
- Costly existing tools or websites in the current market.